Kristian Seegert

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Research Profile

Photonics physicist with expertise in semiconductor photonic devices, nonlinear dynamics, and nanoscale light–matter interaction. My research spans analytical theory, numerical modeling, electromagnetic simulations, device design, and experimental characterization, with a focus on understanding and engineering nanophotonic systems.

Skills

- **Research expertise**
  Semiconductor active devices • Nanophotonics • Non-Hermitian Photonics • Nonlinear dynamics • Nonlinear photonics
- **Methods**
  Quasinormal modes • Coupled-mode theory • Dynamical system theory • k·p theory • Adiabatic approximations
- **Experiments**
  Experimental optics and photonics • Automation of experimental setups • Data-analysis and visualisation
- **Scientific programming**
  MATLAB • Python • COMSOL • Julia
- **Soft skills**
  Presentation of scientific research • Scientific writing • Teaching • Supervision

Education and Research

Technical University of Denmark — **Oct. 2025 – Present**
Postdoctoral researcher, Department of Electrical and Photonics Engineering — *Kgs. Lyngby, Denmark*

Technical University of Denmark — **Mar. 2022 – Nov. 2025**
PhD, Department of Electrical and Photonics Engineering — *Kgs. Lyngby, Denmark*
- **PhD thesis:** Theory and experimental characterization of nanostructured lasers | with Prof. Jesper Mørk

Université Côte d'Azur — **Nov. 2024 – Mar. 2025**
Visiting PhD researcher, INPHYNI — *Nice, France*
- **Project:** Non-Hermitian photonics with coupled nanolasers | with Prof. Fabrice Raineri, Dr. Guilhem Madiot

Technical University of Denmark — **Sep. 2019 – Feb. 2022**
Master of Science in Engineering (Master of Science in Physics and Nanotechnology) — *Kgs. Lyngby, Denmark*
- **Master's thesis:** Theory and Simulation of Semiconductor Fano Lasers | with Prof. Jesper Mørk

Nanyang Technological University — **Sep. 2018 – Dec. 2018**
Exchange semester abroad — *Singapore*

Technical University of Denmark — **Sep. 2016 – Jun. 2019**
Bachelor of Science in Engineering (Bachelor Program in Physics and Nanotechnology) — *Kgs. Lyngby, Denmark*
- **Bachelor's thesis:** Subradiant Multiatom States for Quantum Memory | with Prof. Martijn Wubs

Journal Publications

- **K. Seegert**, R. Gajardo, G. Huyet, F. Raineri & G. Madiot: *Dynamical control of non-Hermitian coupling between sub-threshold nanolasers enables Q-switched pulse generation*, ACS Photonics (in peer review)
- **K. Seegert**, M. Heuck, Y. Yu & J. Mørk: *Instantaneous modes in dispersive laser cavities*, Physical Review A (in peer review)
- J. Mørk, Meng Xiong, **K. Seegert**, M. Marchal, et al.: *Nanostructured Semiconductor Lasers*, IEEE Journal of Selected Topics in Quantum Electronics, **31**, 1 (2025)
- **K. Seegert**, M. Heuck, Y. Yu, & J. Mørk: *Self-pulsing dynamics in lasers with dispersive mirrors*, Physical Review A **109**, 063512 (2024)

- M. V. Gurrieri, E. V. Denning, **K. Seegert**, P. T. Kristensen, & J. Mørk: *Dynamics and condensation of polaritons in an optical nanocavity coupled to two-dimensional materials*, Physical Review B **109**, 155432 (2024)
- M. Rode, G. Meucci, **K. Seegert**, T. Kiørboe, & A. Andersen: *Effects of surface proximity and force orientation on the feeding flows of microorganisms on solid surfaces*, Physical Review Fluids **5**, 123104 (2020).

Conference presentations

- *(Invited)* **K. Seegert**, R. Gajardo (presenter), G. Huyet, F. Raineri & G. Madiot: *Dynamical control of non-Hermitian coupling between sub-threshold nanolasers enables Q-switched pulse generation*, European Optical Society Annual Meeting 2026
- **K. Seegert**, G. Madiot (presenter), G. Beaudoin, K. Pantzas, et al.: *Q-switching with phase-coupled nanolasers*, 33rd Annual Laser Physics Conference 2025
- **K. Seegert**, M. Heuck, Y. Yu, & J. Mørk: *Regimes of self-pulsing in lasers with dispersive mirrors*, The International Symposium on Physics and Applications of Laser Dynamics 2023
- **K. Seegert**, M. Heuck, Y. Yu, & J. Mørk: *Complex dynamics of lasers with passive dispersive reflectors*, Nonlinear Dynamics in Semiconductor Lasers 2023
- **K. Seegert**, Y. Yu, M. Heuck, & J. Mørk: *Dual-mode lasing and beating oscillations in a microscopic laser based on electromagnetically induced transparency*, 2023 Conference on Lasers and Electro-Optics Europe & European Quantum Electronics Conference
- **K. Seegert**, M. Heuck, Y. Yu, & J. Mørk: *Dispersive self-Q-switching in a microscopic laser*, IEEE Photonics Conference 2022

Teaching Experience

DTU Compute, DTU Physics, DTU Electro — **August 2017 – present**

Teaching Assistant at various physics and mathematics courses — *Kgs. Lyngby, Denmark*

- Advanced Engineering Mathematics 1 and 2 (01006, 01035), Physics 1 (10024), Model Physics (10050), Digital Signal Processing (62743), Nanophotonics (34053)

Further Information

Other software: LaTeX • Word • Excel • PowerPoint • GDSPY • Emode Photonix • Inkscape • Maple • Mathematica

Languages: Danish (native) • English (fluent) • Spanish (basic) • German (basic)

Interests: Electronic music • Chess • Philosophy • Sports